**Code No: 126EQ** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year II Semester Examinations, August/September - 2024**
**OBJECT ORIENTED ANALYSIS AND DESIGN**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

Used paper Aug-2024

## PART – A

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | What is a model? | [2] |
| b) | Give example for dependency relationship. | [3] |
| c) | What is an association class? | [2] |
| d) | What are the modeling techniques of package? | [3] |
| e) | What is a swim lane? | [2] |
| f) | What are the different kinds of actions against a message? | [3] |
| g) | What is an event? | [2] |
| h) | How are threads represented in UML diagram? | [3] |
| i) | List artifacts pertaining to elaboration phase. | [2] |
| j) | What type of components are used in unified library applications? | [3] |

## PART – B

**(50 Marks)**

2. Describe the UML (4+1 view) architecture for a software product development. [10]

**OR**

3.a) What are structural things in UML? Explain with suitable diagram.
b) Differentiate between collaboration and use case. [5+5]

4.a) Discuss the adornments for generalization relationship.
b) Demonstrate the use of types and roles in class diagram. [5+5]

**OR**

5. Construct a class diagram and object diagram for school admission system. [10]

6.a) Contrast procedural and nested flow of control.
b) Draw sequence diagrams for a basic flow and an exceptional flow for money transfer using a UPI app. [5+5]

**OR**

7. Describe the common modeling techniques of use case diagrams with suitable diagrams. [10]

8.a) Explain capturing time and space details in a UML diagram.
b) Illustrate history state and composite states. [5+5]

**OR**

9.a) Demonstrate modeling physical aspects of an object-oriented system.
b) What are the common modeling techniques of deployment diagram? [5+5]

10.a) Differentiate between patterns and frameworks.
b) With an example, describe the importance of artifact diagrams. [5+5]

**OR**

11. Demonstrate design view and implementation views of unified library application. [10]

**---ooOoo---**

Used paper Aug-2024